

Lanjing Wang

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EDUCATION

University of Washington PhD in Biomedical Informatics	Sep 2025 - June 2029 Seattle, Washington
The George Washington University PhD student of Biostatistics, Department of Public Health	Aug 2023 – May 2025 Washington, DC
University of California, Berkeley Bachelor of Art in Data Science	Aug 2019 – May 2022 Berkeley, CA

PROFESSIONAL EXPERIENCE

University of Washington

Research Rotation

LLM-based Clinical Literature | Advisor: [Dr. Lucy Lu Wang](#) Sep 2025 – Present

- Develop evaluation benchmarks and LLM-based retrieval methods for clinically relevant literature search in oncology.
- Design structured clinical schemas (e.g., treatment line, biomarkers, and patient context) and guideline-informed search filters to improve retrieval precision.
- Present *LLM-Assisted Generation of Dynamic Shallow Filters from Clinical Guidelines for Oncology Literature Retrieval* at AMIA Conference 2026 Poster presentation.

EHR Research on Epic Cosmos | Advisor: [Dr. Gregory Roth](#) Jan 2026 – Present

- Develop scalable EHR infra with an agent-based querying layer for cohort construction and longitudinal analysis.
- Quantify adoption of guideline-recommended treatment strategies and the temporal lag from clinical evidence to real-world practice using EPIC Cosmos data.

Hope AI

Jan 2025 - April 2026

Biomedical Data Science Researcher, Advisor: Dr. Will Ma and Dr. Feifang Hu

- Developed end-to-end clinical LLM pipelines for large-scale clinical trial literature retrieval and evidence extraction, transforming unstructured literature into structured analysis-ready datasets for evidence synthesis, subgroup analysis, pooling analysis and meta-analysis on Oncology diseases.
- Designed agentic literature review workflows integrating study screening, clinical context schema design, and automated statistical analysis to support multi-source clinical evidence synthesis; manuscript currently in preparation for submission.
- Built LLM-based algorithms on large-scale EHR datasets from Mayo Clinic to transform unstructured clinical notes and EHR data into structured clinical outcomes (e.g., OS, PFS) to support evidence comparison between clinical trial literature and real-world EHR data.

George Washington University Biostatistics Center

Aug 2024 – July 2025

Senior Research Assistant, Advisor: Dr. Guoqing Diao

- Applied statistical methods for a large-scale systematic review on the epidemiology of pulmonary hypertension in sub-Saharan Africa, including meta-analysis, meta-regression, missing data imputation, and sensitivity analysis across multi-study clinical datasets.
- Conducted pooled statistical analyses and subgroup heterogeneity analyses across HIV, sickle cell disease, and rheumatic heart disease populations to support epidemiological modeling, evidence synthesis, and manuscript preparation for publication.

American Heart Association

Data Science Intern

Feb 2022 – May 2023

Data Science Fellow, Advisor: Dr. Jennifer Hall and Dr. Xue Feng

July 2022 – Aug 2023

- Data Science: GWTG registry data: Data Preprocessing and Database Design on [Precision Medicine Platform](#)

- Led and designed the data preprocessing automated pipelines in Python for over +13 million registry records, with the goal of cleaning, transforming, and integrating medical data to prepare them for data quality control and data analysis.
 - Developed a cloud-based database platform system utilizing the Python-based Django web framework to host and serve more than 13 million medical data records on the AWS servers.
2. Clinical Research
- Worked with clinical research to conduct statistical analyses and established hypothesis tests to evaluate in-hospital mortality in STEMI patients and assessed the impact of STEMI patients based on pre-pandemic and COVID-19 pandemic trends, using data from the GWTG-CAD registry.
3. Periodic Table of Food Initiative (PTFI): Food Research Database Design
- Designed and maintained data pipelines to collect, clean, and integrate food-related datasets into the database, collaborating with food researchers to ensure data accuracy and accessibility for research and analysis.
4. Broad Institute: Genomics Platform Evaluation [AnVIL \(Terra\)](#)
- Collaborated with clinical researchers to assess the capabilities, limitations, and performance of genomics platforms.

Akai Technology Co., Ltd.

Dec 2020 – Mar 2021

Machine Learning Intern

- Developed biological cell counting recognition algorithms by building neural network models using Faster R-CNN, Mask R-CNN for image recognition and detection ([Demo](#)).

ByteDance - TikTok

May 2021 – Aug 2021

TikTok Overseas e-commerce – Data Analyst Intern

- Analyzed product datasets and sorted out 300+ potential hot-selling products for the sales direction of e-business live broadcast in the Indonesian market. Successfully raised sales by 50% and generated 10k+ impressions on TikTok.

Ernst & Young (EY)

Sep 2020 – Dec 2020

Consulting Department – Business Analyst Intern

- Designed a robotic process automation (RPA) evaluation matrix to help clients make data-driven decisions and visualize process prioritization; built RPA solutions and code based on user requirements, workflows, and pain points.

JD.com

Mar 2021 – May 2021

Data Analyst Intern

- Monitored and Managed data operations for 24 warehouses across 7 cities; built Excel/VBA pipelines for daily/weekly/monthly reporting and partnered with 20 departments to fix data-system gaps.

PUBLICATIONS

1. Beon C, **Wang L**, Manchanda V, Mallya P, Hong H, Picotte H, Thomas K, Hall JL, Zhao J, Feng X. Empowering Research With the American Heart Association Get With The Guidelines® Registries Through Integration of a Database and Research Tools. *Circulation. Cardiovascular quality and outcomes*, 2024 Sep;17(9):e010967. [doi: 10.1161/CIRCOUTCOMES.124.010967](https://doi.org/10.1161/CIRCOUTCOMES.124.010967)
2. **Wang L**, Manchanda V, Picotte H, Beon C, Hall JL, Zhao J, Feng X. Synthetic Data for the Get With The Guidelines® Stroke Registry. *Oral presentation at AHA Scientific sessions 2023*. doi.org/10.1161/JAHA.124.039667
3. Zègre-Hemsey J, Goyal A, Poudel R, Thomas K, Ali M, Best P, Bieniarz M, Fonarow G, French W, Granger C, Henry T, Hong H, Jollis J, Redlener M, Spier T, Stone H, Wahab F, **Wang L**, Jacobs A. Treatment Times and In-Hospital Mortality Among Patients with ST-Elevation Myocardial Infarction Throughout the Waves of the COVID-19 Pandemic: Lessons Learned. *COVID* 2025, 5, 114. <https://doi.org/10.3390/covid5080114>
4. Zeder K; **Wang L**; Ngah V; Santi A; Robbins E; Mocumbi A; Namuyonga J; Nyasulu P; Kovacs G; Humbert M; Strange G; Stewart S; Diao G; Maron B. The Epidemiology of Pulmonary Hypertension in sub-Saharan Africa – A systematic report (*under review at Wiley Pulmonary Circulation*)
5. Zhao Z; Li Q; Wu Y; **Wang L**; Shi J; Xu Y; Xie E; Luo R; Ren Z. Systematic Literature Reviews With Two Multi-Agent Systems And Human-In-The-Loop (*under review at STAI-X 2026*)

ABSTRACTS

1. **Wang L**, Lucy W, LLM-Assisted Generation of Dynamic Shallow Filters from Clinical Guidelines for Oncology Literature Retrieval. *Poster Presentation at AMIA Conference 2026*
2. Zeder K, **Wang L**, Santi A, Ngah V, Robbins E, Diao G, Maron B. The Prevalence of Pulmonary Hypertension in HIV in sub-Saharan Africa: A Systematic Review and Meta-Analysis. *Oral presentation at PVRI conference 2025*
3. Santi A, **Wang L**, Ngah V, Robbins E, Diao G, Maron B, Zeder K. The Prevalence of Pulmonary Hypertension in Rheumatic Heart Disease/Sickle Cell Disease in sub-Saharan Africa: A Systematic Review and Meta-Analysis. *Oral presentation at PVRI conference 2025*
4. Ngah V, **Wang L**, Santi A, Robbins E, Diao G, Maron B, Zeder K. The Prevalence of Pulmonary Hypertension in Sickle Cell Disease in sub-Saharan Africa: A Systematic Review and Meta-Analysis. *Poster presentation at European Respiratory Society Conference (ERS) 2025*

SKILLS

- Programming Languages: Python (django, cv2, scikit-learn, tensorflow, pandas, numpy), R (mice, tidyverse, meta, dplyr, table1), SAS, SQL, Git/Github
- Hobbies: flying Cessna airplanes (Private Pilot License in progress), cooking, coding